AKSHAYA S

192412363

ECA5601

5G/ 6G COMMUNICATION SYSTEM

Ex.No. 13: V2X Latency Comparison

AIM:

To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Procedure :

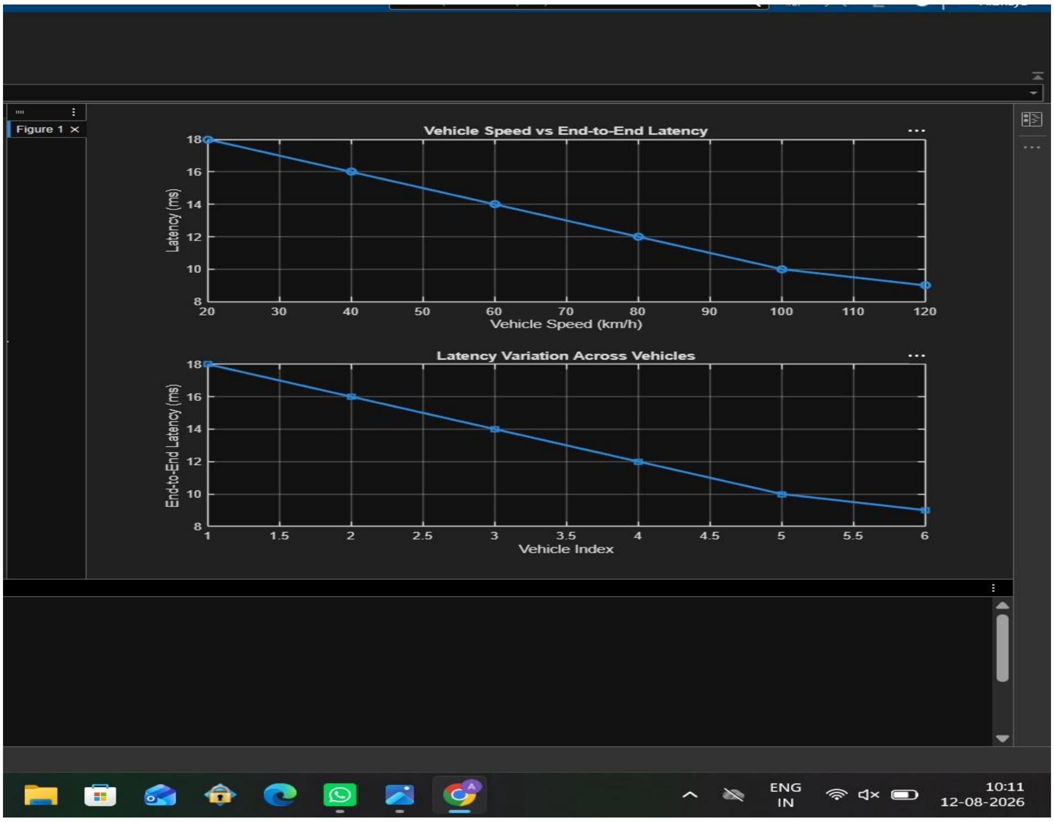
1. Define the V2X communication parameters such as vehicle speed, transmission range, packet size, and simulation time.
2. Generate different vehicle movement scenarios by varying the vehicle speed.
3. Simulate wireless communication between vehicles using a V2X communication model.
4. Calculate the End-to-End Latency (ms) for data transmission between communicating vehicles.
5. Compute the Packet Delivery Ratio (PDR %) by comparing the number of successfully received packets with transmitted packets.
6. Analyze the effect of increasing Vehicle Speed (km/h) on communication latency and packet delivery performance.
7. Plot MATLAB graphs for End-to-End Latency, Packet Delivery Ratio, and Vehicle Speed.
8. Compare the obtained results to evaluate the efficiency and reliability of the V2X communication system under different traffic conditions.

Objective 1 : To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.

Program :

```
clc; clear;
close all;
% Vehicle Speed (km/h) speed
= 20:20:120;
% Simulated End-to-End Latency (ms) latency = [18
16 14 12 10 9]; disp('Vehicle Speed (km/h) End-to-
End Latency (ms)') disp([speed' latency']) figure
% -------- Graph 1 -------- subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)') grid on
% -------- Graph 2 -------- subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index') ylabel('End-to-End
Latency (ms)') grid on
```

Output :



Objective 2 : To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.

Program :

clc; clear;

close all;

% Vehicle Speed (km/h) speed

= 20:20:120;

% Simulated Packet Delivery Ratio (%) PDR = [99 98

97 95 93 90]; disp('Vehicle Speed (km/h) Packet

Delivery Ratio (%)') disp([speed' PDR']) figure

% -------- Graph 1 --------
subplot(2,1,1) plot(speed,PDR,'-

o','LineWidth',2) title('Vehicle Speed vs

Packet Delivery Ratio') xlabel('Vehicle Speed (km/h)') ylabel('Packet Delivery Ratio (%)')

grid on

% -------- Graph 2 -------- subplot(2,1,2)

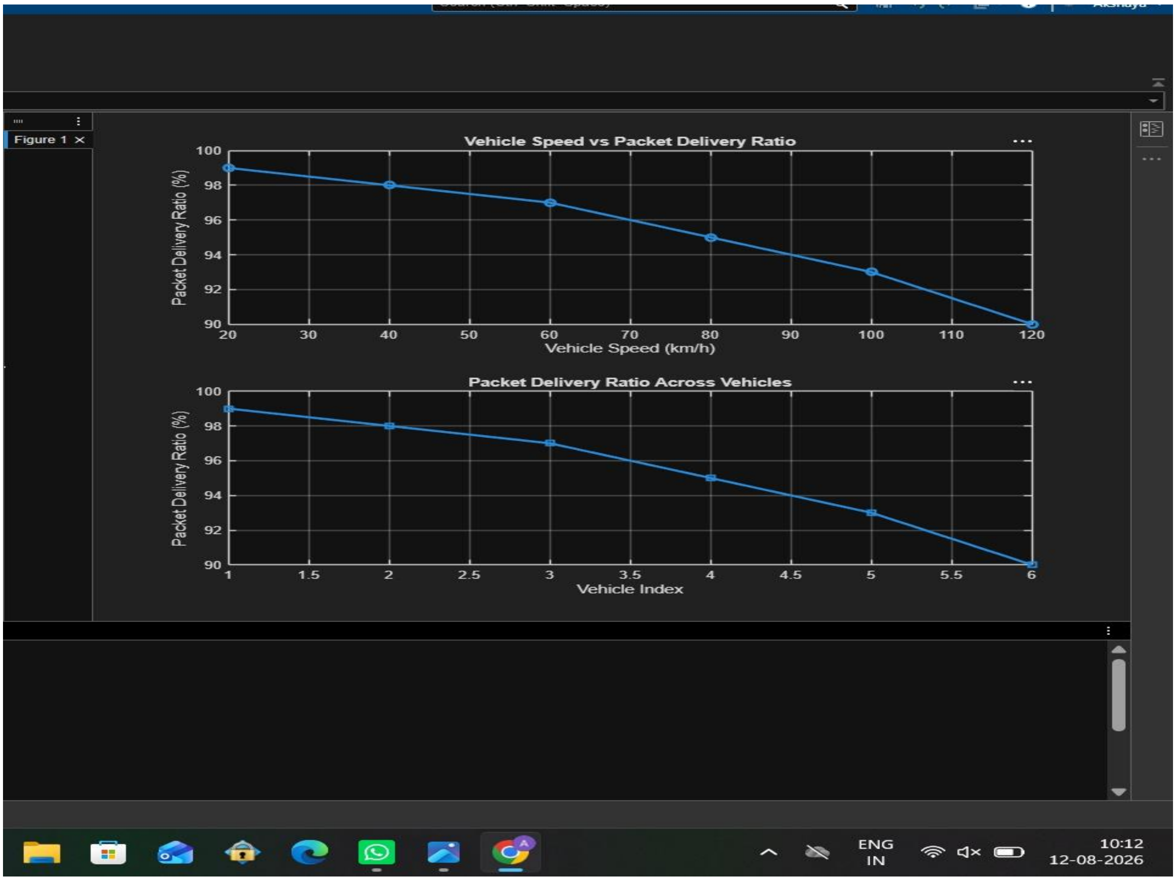
plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index') ylabel('Packet

Delivery Ratio (%)') grid on

Output :



Objective 3 : To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

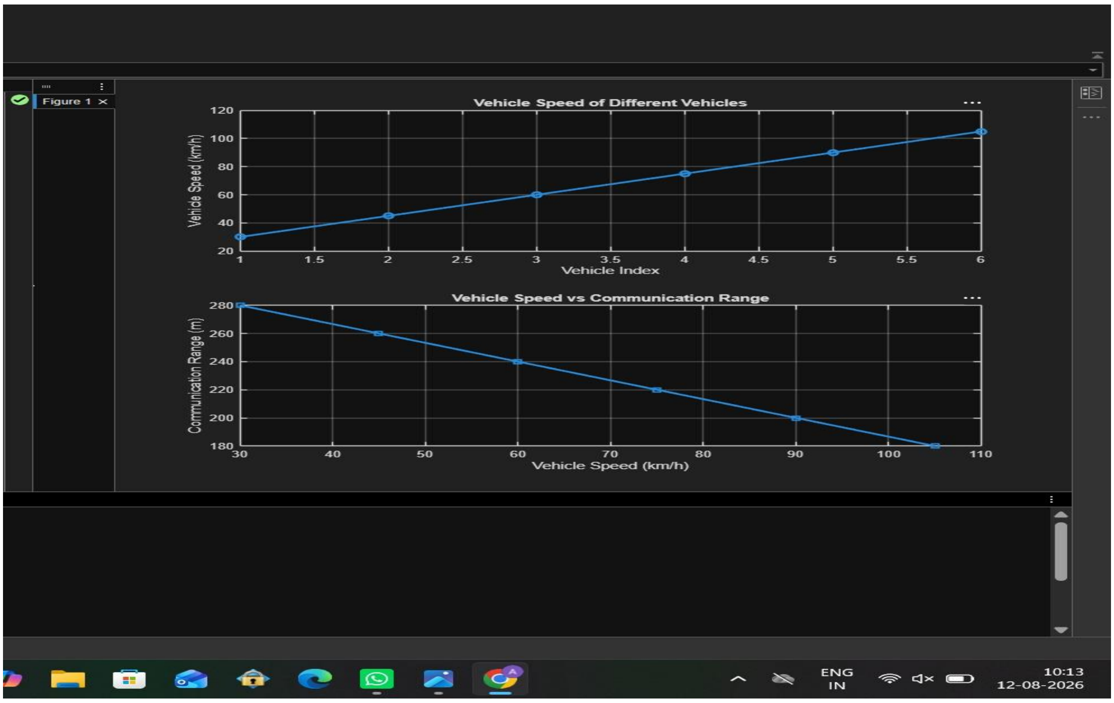
Program :

clc; clear;

close all;

```
% Vehicle IDs vehicle = 1:6; % Vehicle Speed (km/h)
speed = [30 45 60 75 90 105]; % Communication
Range (m) range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range']) figure
% -------- Graph 1 -------- subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index') ylabel('Vehicle
Speed (km/h)') grid on
% -------- Graph 2 -------- subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)') grid on
```

Output :



Result:

1.The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.

2.The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.

3.The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.